

Name: Patrick salazar

Bears, Species and DNA

~home edition

Go to [Bears, Species & DNA](#) and use the paw to move forward in the lesson

1. What do the nodes on the tree diagram represent?

It represents the species lineages

2. Trees are made using evidence from what four areas?

Anatomy, behavior, geography, and fossils

3. What bear is most closely related to the brown bear?
Why is determining the timing of the split between these two species difficult?

The polar bear. Determining is difficult Because of gaps or conflicting evidence

4. How can DNA be used to determine the timing of divergence?

They looked back at when the new dna version appeared

5. What types of genes change very slowly?

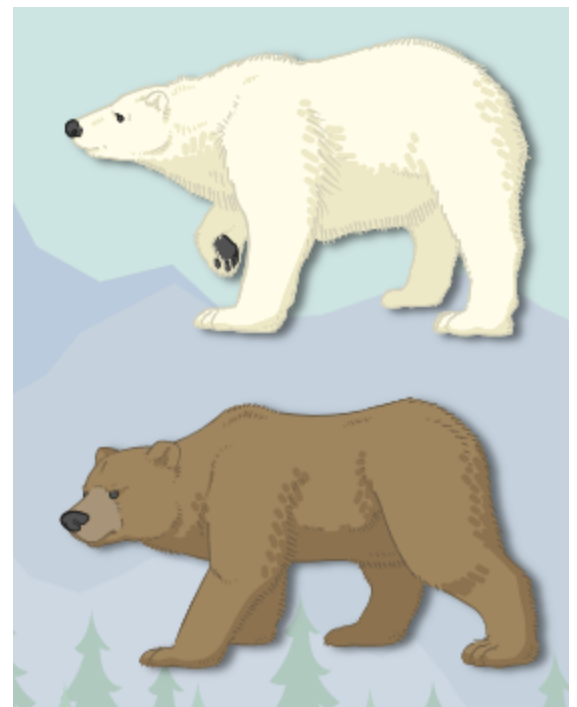
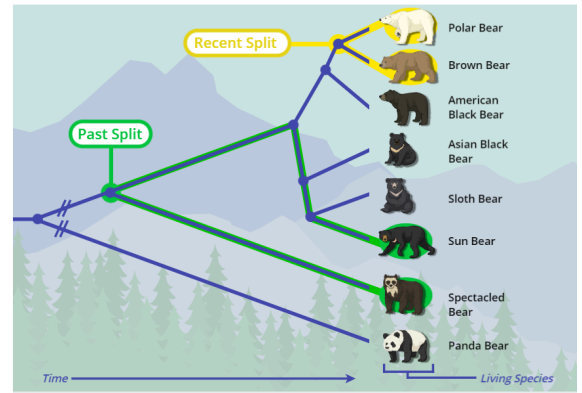
Genes that code for proteins

What types of genes might change quickly?

Genes that influence traits

6. What is a cladogram? What can it tell you?

It is a gene tree and it can tell you an accurate timing of a species split

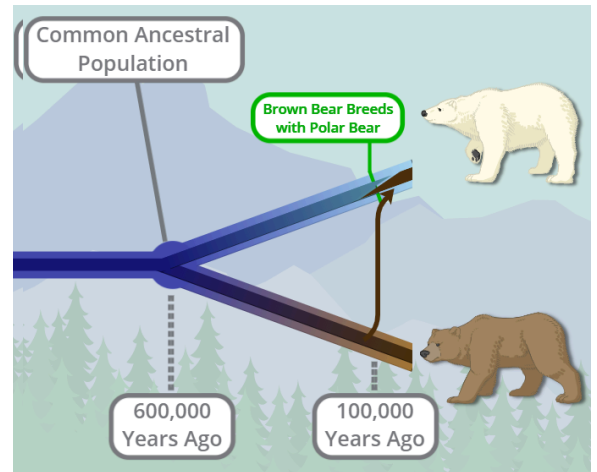


7. Nuclear DNA is not the only way to construct a tree, what other source of DNA can be used?

Mitochondria dna

8. Why does the mitochondrial gene tree suggest a later split? Why the disparity between the nuclear gene tree?

It only holds a maternal side



9. Why is it important to look at multiple types of evidence to understand a species' evolutionary history?

Because it can show when the split was and show more detailed answers than the other one

Part 2: Why is this information important?

Understanding evolutionary relationships in animals has some implications for conservation and saving endangered species. Consider the story about Florida panthers

In 1995, the Florida panther was on the brink of extinction, there were only an estimated 30 panthers left and inbreeding has resulted in many genetic defects. Florida officials decided to save the animal by bringing to Florida eight Texas cougars. The move was not without criticism; the cougar and panther are not the same species and some argued that this wouldn't save the panther, but would instead just create a new type of big cat. Recent studies suggest the program was a success, as it is now estimated that 200 panthers now roam Florida.



10. Climate change has scientists worried about the future of the polar bear. What lesson could we take from Florida's effort to save the panther to apply to future conservation efforts with the polar bear?

I think breeding with an animal with similar genes could maybe help. I would go with a brown bear because they were the closest relative of the polar bear.

